

Submit 8 problems total from the union of *LADW* Ch 1.5-7 & Ch 2.1-3 (copied below) and the supplementary problems.

Please label all *LADW* problems by “Chapter.Section.Exercise,” e.g. 1.3.4. Label the supplementary problems as they’re labeled on the handout. Please make a note about any collaboration or use of outside resources.

Please put a “G” in front of the four problems you would like graded for correctness. Additionally, put an “F” in front of any of the problems graded for completion on which you would like written feedback from the grader.

Suggested *LADW* problems: 1.5.5, 1.6.5, 1.6.9, 1.7.1, 1.7.5, 2.2.2, 2.3.4, and 2.3.6

Section 1.5

Exercises.

X

5.1. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & -2 \end{pmatrix}, C = \begin{pmatrix} 1 & -2 & 3 \\ -2 & 1 & -1 \end{pmatrix}, D = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$$

- Mark all the products that are defined, and give the dimensions of the result: $AB, BA, ABC, ABD, BC, BC^T, B^TC, DC, D^TC^T$.
- Compute $AB, A(3B + C), B^TA, A(BD), (AB)D$.

5.2. Let T_γ be the matrix of rotation by γ in $\mathbb{R}^2$. Check by matrix multiplication that $T_\gamma T_{-\gamma} = T_{-\gamma} T_\gamma = I$.

5.3. Multiply two rotation matrices T_α and T_β (it is a rare case when the multiplication is commutative, i.e. $T_\alpha T_\beta = T_\beta T_\alpha$, so the order is not essential). Deduce formulas for $\sin(\alpha + \beta)$ and $\cos(\alpha + \beta)$ from here.

5.4. Find the matrix of the orthogonal projection in $\mathbb{R}^2$ onto the line $x_1 = -2x_2$. **Hint:** What is the matrix of the projection onto the coordinate axis x_1 ?

5.5. Find linear transformations $A, B: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $AB = \mathbf{0}$ but $BA \neq \mathbf{0}$.

5.6. Prove Theorem 5.1, i.e. prove that $\text{trace}(AB) = \text{trace}(BA)$.

5.7. Construct a non-zero matrix A such that $A^2 = \mathbf{0}$.

5.8. Find the matrix of the reflection through the line $y = -2x/3$. Perform all the multiplications.

Section 1.6

Exercises.

6.1. Prove, that if $A: V \rightarrow W$ is an isomorphism (i.e. an invertible linear transformation) and $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis in V , then $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n$ is a basis in W .

6.2. Find all right inverses to the 1×2 matrix (row) $A = (1, 1)$. Conclude from here that the row A is not left invertible.

6.3. Find all left inverses of the column $(1, 2, 3)^T$.

6.4. Is the column $(1, 2, 3)^T$ right invertible? Justify

6.5. Find two matrices A and B that AB is invertible, but A and B are not. **Hint:** square matrices A and B would not work. **Remark:** It is easy to construct such A and B in the case when AB is a 1×1 matrix (a scalar). But can you get 2×2 matrix AB ? 3×3 ? $n \times n$?

6.6. Suppose the product AB is invertible. Show that A is right invertible and B is left invertible. **Hint:** you can just write formulas for right and left inverses.

6.7. Let A and AB be invertible (assuming that the product AB is well defined). Prove that B is invertible.

6.8. Let A be $n \times n$ matrix. Prove that if $A^2 = \mathbf{0}$ then A is not invertible

6.9. Suppose $AB = \mathbf{0}$ for some non-zero matrix B . Can A be invertible? Justify.

6.10. Write matrices of the linear transformations T_1 and T_2 in $\mathbb{R}^5$, defined as follows: T_1 interchanges the coordinates x_2 and x_4 of the vector $\mathbf{x}$, and T_2 just adds to the coordinate x_2 a times the coordinate x_4 , and does not change other coordinates, i.e.

$$T_1 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_4 \\ x_3 \\ x_2 \\ x_5 \end{pmatrix}, \quad T_2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 + ax_4 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix};$$

here a is some fixed number.

Show that T_1 and T_2 are invertible transformations, and write the matrices of the inverses. **Hint:** it may be simpler, if you first describe the inverse transformation, and then find its matrix, rather than trying to guess (or compute) the inverses of the matrices T_1, T_2 .

6.11. Find the matrix of the rotation in $\mathbb{R}^3$ through the angle α around the vector $(1, 2, 3)^T$. We assume that the rotation is counterclockwise if we sit at the tip of the vector and looking at the origin.

You can present the answer as a product of several matrices: you don’t have to perform the multiplication.

6.12. Give examples of matrices (say 2×2) such that:

- $A + B$ is not invertible although both A and B are invertible;
- $A + B$ is invertible although both A and B are not invertible;
- All of A, B and $A + B$ are invertible

6.13. Let A be an invertible symmetric ($A^T = A$) matrix. Is the inverse of A symmetric? Justify.

Section 1.7

Exercises.

7.1. Let X and Y be subspaces of a vector space V . Prove that $X \cap Y$ is a subspace of V .

7.2. Let V be a vector space. For $X, Y \subset V$ the sum $X + Y$ is the collection of all vectors $\mathbf{v}$ which can be represented as $\mathbf{v} = \mathbf{x} + \mathbf{y}$, $\mathbf{x} \in X, \mathbf{y} \in Y$. Show that if X and Y are subspaces of V , then $X + Y$ is also a subspace.

7.3. Let X be a subspace of a vector space V , and let $\mathbf{v} \in V, \mathbf{v} \notin X$. Prove that if $\mathbf{x} \in X$ then $\mathbf{x} + \mathbf{v} \notin X$.

7.4. Let X and Y be subspaces of a vector space V . Using the previous exercise, show that $X \cup Y$ is a subspace if and only if $X \subset Y$ or $Y \subset X$.

7.5. What is the smallest subspace of the space of 4×4 matrices which contains all upper triangular matrices ($a_{j,k} = 0$ for all $j > k$), and all symmetric matrices ($A = A^T$)? What is the largest subspace contained in both of those subspaces?

Section 2.1

No exercises in this section.

Section 2.2

Exercises.

X

2.1. Write the systems of equations below in matrix form and as vector equations:

$$\begin{aligned} \text{a) } & \begin{cases} x_1 + 2x_2 - x_3 = -1 \\ 2x_1 + 2x_2 + x_3 = 1 \\ 3x_1 + 5x_2 - 2x_3 = -1 \end{cases} \\ \text{b) } & \begin{cases} x_1 - 2x_2 - x_3 = 1 \\ 2x_1 - 3x_2 + x_3 = 6 \\ 3x_1 - 5x_2 = 7 \\ x_1 + 5x_3 = 9 \end{cases} \\ \text{c) } & \begin{cases} x_1 + 2x_2 + 2x_4 = 6 \\ 3x_1 + 5x_2 - x_3 + 6x_4 = 17 \\ 2x_1 + 4x_2 + x_3 + 2x_4 = 12 \\ 2x_1 - 7x_3 + 11x_4 = 7 \end{cases} \\ \text{d) } & \begin{cases} x_1 - 4x_2 - x_3 + x_4 = 3 \\ 2x_1 - 8x_2 + x_3 - 4x_4 = 9 \\ -x_1 + 4x_2 - 2x_3 + 5x_4 = -6 \end{cases} \end{aligned}$$

$$\begin{aligned} \text{e)} & \begin{cases} x_1 + 2x_2 - x_3 + 3x_4 = 2 \\ 2x_1 + 4x_2 - x_3 + 6x_4 = 5 \\ + 2x_4 = 3 \end{cases} \\ \text{f)} & \begin{cases} 2x_1 - 2x_2 - x_3 + 6x_4 - 2x_5 = 1 \\ x_1 - x_2 + x_3 + 2x_4 - x_5 = 2 \\ 4x_1 - 4x_2 + 5x_3 + 7x_4 - x_5 = 6 \end{cases} \\ \text{g)} & \begin{cases} 3x_1 - x_2 + x_3 - x_4 + 2x_5 = 5 \\ x_1 - x_2 - x_3 - 2x_4 - x_5 = 2 \\ 5x_1 - 2x_2 + x_3 - 3x_4 + 3x_5 = 10 \\ 2x_1 - x_2 - 2x_4 + x_5 = 5 \end{cases} \end{aligned}$$

Solve the systems. Write the answers in the vector form.

2.2. Find all solutions of the vector equation

$$x_1 \mathbf{v}_1 + x_2 \mathbf{v}_2 + x_3 \mathbf{v}_3 = \mathbf{0},$$

where $\mathbf{v}_1 = (1, 1, 0)^T$, $\mathbf{v}_2 = (0, 1, 1)^T$ and $\mathbf{v}_3 = (1, 0, 1)^T$. What conclusion can you make about linear independence (dependence) of the system of vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$?

Section 2.3

Exercises.

3.1. For what value of b the system

$$\begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 4 \\ b \end{pmatrix}$$

has a solution. Find the general solution of the system for this value of b .

3.2. Determine, if the vectors

$$\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

are linearly independent or not.

Do these four vectors span $\mathbb{R}^4$? (In other words, is it a generating system?) What about $\mathbb{C}^4$?

X 3.3. Determine, which of the following systems of vectors are bases in $\mathbb{R}^3$:

- $(1, 2, -1)^T, (1, 0, 2)^T, (2, 1, 1)^T$;
- $(-1, 3, 2)^T, (-3, 1, 3)^T, (2, 10, 2)^T$;
- $(67, 13, -47)^T, (\pi, -7.84, 0)^T, (3, 0, 0)^T$.

Which of the systems are bases in $\mathbb{C}^3$?

3.4. Do the polynomials $x^3 + 2x$, $x^2 + x + 1$, $x^3 + 5$ generate (span) $\mathbb{P}_3$? Justify your answer.

3.5. Can 5 vectors in $\mathbb{F}^4$ be linearly independent? Justify your answer.

3.6. Prove or disprove: If the columns of a square $(n \times n)$ matrix A are linearly independent, so are the columns of $A^2 = AA$.

3.7. Prove or disprove: If the columns of a square $(n \times n)$ matrix A are linearly independent, so are the rows of $A^3 = AAA$.

3.8. Show that if the equation $A\mathbf{x} = \mathbf{0}$ has unique solution (i.e. if echelon form of A has pivot in every column), then A is left invertible. **Hint:** elementary matrices may help.

Note: It was shown in the text that if A is left invertible, then the equation $A\mathbf{x} = \mathbf{0}$ has unique solution. But here you are asked to prove the converse of this statement, which was not proved.

Remark: This can be a very hard problem, for it requires deep understanding of the subject. However, when you understand what to do, the problem becomes almost trivial.

3.9. Is the reduced echelon form of a matrix unique? Justify your conclusion.

Namely, suppose that by performing some row operations (not necessarily following any algorithm) we end up with a reduced echelon matrix. Do we always end up with the same matrix, or can we get different ones? Note that we are only allowed to perform row operations, the "column operations" are forbidden.

Hint: What happens if you start with an invertible matrix? Also, are the pivots always in the same columns, or can it be different depending on what row operations you perform? If you can tell what the pivot columns are without reverting to row operations, then the positions of pivot columns do not depend on them.

**Note: we will discuss and prove
Exercise 2.3.9 in class.**

Supplementary problems for HW2

I. a) Give an example of an invertible matrix in $M_{2 \times 2}^{\mathbb{Z}/5\mathbb{Z}}$ and also give its inverse. Then given an example of a non-invertible matrix in $M_{2 \times 2}^{\mathbb{Z}/5\mathbb{Z}}$ and explain why it is not invertible.

b) How many different linear transformations are there from $(\mathbb{Z}/5\mathbb{Z})^2$ to $(\mathbb{Z}/5\mathbb{Z})^2$? How many of these are invertible? Explain your calculations.

II. How many subspaces does $(\mathbb{Z}/5\mathbb{Z})^2$ have? What about $(\mathbb{Z}/5\mathbb{Z})^3$? $(\mathbb{Z}/p\mathbb{Z})^n$? Explain your calculations.

III. Solve the following two system of linear equations. The first is over the field $\mathbb{Z}/5\mathbb{Z}$, the second is over $\mathbb{C}$. Write your answers in vector form.

$$\begin{aligned} 2x_1 + 3x_2 + x_3 + x_4 &= 0 \\ 3x_1 + x_2 + x_3 + x_4 &= 0 \\ x_1 + 3x_2 + 3x_3 + x_4 &= 0 \end{aligned}$$

$$\begin{aligned} x_1 + ix_2 + (1+i)x_3 &= 1 \\ 3x_1 - ix_2 + x_3 &= i \\ ix_1 + 2x_2 + (2-i)x_3 &= -1 \end{aligned}$$

IV. Write each of these matrices as a product of elementary matrices.

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 5 & 1 \\ 2 & 11 & 5 \\ 0 & 1 & 0 \end{pmatrix}$$

V. Find all real polynomials $p(x)$ of degree at most 3 such that $p(x) = p'(x)$ when $x = 0$ and when $x = 1$.